# Juan Antonio Monleón de la Lluvia

Nuclear Engineer - PhD Student

## CONTACT


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## EDUCATION

**Double Master in Industrial Engineering and Nuclear Science and Technology**
Polytechnic University of Madrid — 2021-2023

**Bachelor in Industrial Technology Engineering**
Polytechnic University of Madrid — 2017-2021

## TECHNICAL SKILLS

**Nuclear Codes:** MCNP, Serpent, OpenMC, NJOY, ADVANTG

**Programming:** Python, Git, VS Code, Claude Code

**Web / App Dev:** React, TypeScript, Tauri, FastAPI

**Documentation:** LaTeX, Microsoft 365

## LANGUAGES

**Spanish** Native language

**English** C2 Level

**French** B2 Level

## ABOUT ME

I am a passionate professional dedicated to the nuclear sector, driven by a strong interest in research and innovation. As a proactive person, I constantly seek out new learning opportunities to expand my knowledge and skills. I enjoy applying my programming expertise to solve complex problems and enhance both my daily work and personal projects.

## RESEARCH EXPERIENCE

### PhD in Neutronics

ASNR (formerly IRSN), France — Nov 2023 - Present

- Propagating nuclear-data uncertainties through Monte Carlo neutronic simulations of PWR vessel ageing via sensitivity analysis and Monte Carlo sampling, including the re-evaluation of $^{56}$Fe elastic angular distributions and covariances from EXFOR experimental data
- Applying variance-reduction techniques (ADVANTG weight windows) to improve computational efficiency in deep-penetration shielding calculations
- Supervised a Master's student (Mar–Sep 2025) on nuclear-data uncertainty quantification in source-term calculations of PWRs

### Research Internship

SCK CEN, Belgium — Sep 2022 - Jun 2023

- Completed Master's thesis titled **"Reducing Spectrum-Driven Uncertainties with Variance Reduction Techniques"**
- Conducted neutronic and deep shielding analyses using MCNP and ADVANTG for the MYRRHA Project

### Master Thesis Research

UPM, Spain — Jun 2023 - Oct 2023

- Developed **"Enhancing the EXFOR nuclear data library with Machine Learning Techniques"**
- Implemented outlier detection methods for proton-induced reactions

### Master Internship

ENUSA Advanced Industries, Spain — Oct 2021 - Apr 2022

- Designed and developed a Python program for the analysis and inspection of fuel assemblies and spent fuel

## PUBLICATIONS

## AWARDS

**Best Academic Report for Nuclear Master**

Polytechnic University of Madrid (2022-2023)

**Best Master Thesis Award**

SCK CEN, Belgium (2023)

**Best Poster Award**

Journées des Thèses of IRSN (2024) - **View Poster**

## COMMUNITY

**JEFF Community**

Active member — Joint Evaluated Fission and Fusion File (OECD/NEA)

**SINBAD Community**

Member — Shielding Integral Benchmark Archive & Database (OECD/NEA)

- **J. A. Monleon de la Lluvia, M. Brovchenko, D. Rochman, E. Dumonteil**, **"Towards Efficient Nuclear Data Uncertainty Quantification in Radiation Shielding Calculations"**, *Nuclear Science and Engineering*, 2025. DOI: 10.1080/00295639.2025.2510048 · **pre-print** — 2025

## CONFERENCES & PRESENTATIONS

### FIRST AUTHOR

- **"Nuclear Data Uncertainty Quantification for PWR Pressure Vessel Shielding Calculations"** — RPSD 2026, South Korea *(accepted)* — Oct 2026
- **"KIKA: A Unified Desktop Application for Nuclear Data Processing, Sensitivity, and Uncertainty Quantification"** — RPSD 2026, South Korea *(accepted)* — Oct 2026
- **"Evaluation of elastic angular distribution covariances for $^{56}$Fe from EXFOR experimental data"** — WONDER 2026, Aix-en-Provence *(upcoming)* — Jun 2026
- **"Impact of Elastic Scattering Angular Distribution Covariances on Fast Neutron Fluence; Application to $^{56}$Fe"** — PHYSOR 2026 — Apr 2026
- **"Sensitivity Analysis and Uncertainty Quantification in PWR Irradiation Ageing-like problems"** — ANS Winter Conference RPSD, Orlando — Nov 2024

### CO-AUTHORED

- *Co-author:* "Sensitivity Coefficients of a Single Response Function in Fixed-Source Calculations Using SERPENT2 and MCNP6" — PHYSOR 2026 — Apr 2026
- *Co-author:* "Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations" — PHYSOR 2026 — Apr 2026

## SOFTWARE & TOOLS

### KIKA — Nuclear-data Desktop Application

Lead developer — 2025 - Present

- Cross-platform desktop app (Windows / macOS / Linux) for nuclear-data visualization, sensitivity analysis, and uncertainty quantification
- Tauri + React + TypeScript frontend with a FastAPI + Python backend; built on top of **kika-nd**, the open-source Python library (PyPI) providing the computational core (ENDF / ACE covariances, MCNP interop)

## TRAINING

- OpenMC Course — NEA, Paris — Dec 2025
- Master in Management and Direction of SMEs | EFEM — 2024-2025
- NJOY Course - NEA, Paris — Dec 2024
- Sensitivity Analysis Summer School (SAMO) | UNIPR, Parma — Jun 2024
- Serpent Bootcamp | NEA, Paris — Nov 2023
- Course in "Nuclear Data for Energy and Non-Energy Applications" | GREAT PIONEER — Dec 2022

## REFERENCES

**Dr. Mariya Brovchenko**
ASNR, France
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